

Parts Receiving & Stock Transfer Process

Global ERP -- Workshop Parts Management

Process Overview

All parts received from vendors enter through the **Main Warehouse** (central receiving point), then are transferred to the **Workshop Branch** location where the car is being serviced.



Step 1: Goods Receipt (GRN)

Purpose

Record the physical receipt of parts from vendor into the Main Warehouse.

Trigger

- Workshop staff confirms delivery against Purchase Order (PO)

Endpoint

POST /api/company/{companyId}/workshop/procurement/{poId}/receive

Input

Field	Required	Description
items[].itemId	Yes	Purchase order item UUID
items[].receivedQty	Yes	Quantity physically received
grnNumber	No	Auto-generated if not provided

Process

- 1. Validate PO exists and is not cancelled
- 2. Validate each item belongs to this PO
- 3. Validate received qty does not exceed ordered qty
- 4. Update PO item: `received_qty += deliveredQty`
- 5. Update PO item status: `received` (if full) or `partial`
- 6. Update PO status: `received` (if all items done) or `partially_received`
- 7. Create inventory movement record (GRN)
- 8. Update inventory stock at Main Warehouse
- 9. Create accounting journal entry
- 10. Update part quote status

Database Changes

purchase_order_items:

Column	Change
received_qty	<code>+= delivered quantity</code>
status	<code>received</code> (if full) or <code>partial</code>

purchase_orders:

Column	Change
status	<code>received</code> (all done) or <code>partially_received</code>

inventory_movements:

Column	Value
company_id	Company UUID
source_type	<code>receipt</code>
source_id	PO item UUID
movement_type	<code>in</code>
quantity	Received qty
grn_number	GRN-{pold}-{itemId}
purchase_order_id	PO UUID
location_id	Main Warehouse location UUID

inventory_stock:

Column	Change
location_id	Main Warehouse

Column	Change
quantity	$\text{+= received qty (UPSERT)}$

accounting_journals:

Column	Value
reference	GRN- $\{\text{pold}\}$ - $\{\text{itemId}\}$
description	Goods Receipt for PO $\{\text{poNumber}\}$
is_posted	true

accounting_journal_lines:

Line	Account	Debit	Credit
1	Inventory (Asset)	amount	0
2	Accounts Payable (Liability)	0	amount

$\text{amount} = \text{received_qty} \times \text{unit_cost}$

part_quotes:

Column	Change
status	Received
delivery_note_status	received

Cases

Case	Behavior	PO Item Status	PO Status
Full delivery (all items, all qty)	All items received	received	received
Partial delivery (some items only)	Only delivered items updated	received / unchanged	partially_received
Partial quantity (2 of 5 ordered)	$\text{received_qty} = 2$	partial	partially_received
Second delivery (remaining 3)	$\text{received_qty} = 5$	received	received (if all done)
Over-delivery (6 of 5 ordered)	REJECTED	No change	No change
PO cancelled	BLOCKED	No change	No change
PO already fully received	BLOCKED	No change	No change

Case	Behavior	PO Item Status	PO Status
Zero cost item	Receipt OK, no journal entry	Updated	Updated
No accounting entity configured	Receipt OK, journal skipped	Updated	Updated

Accounting Impact

Balance Sheet:

Assets:	Inventory	+amount (debit)
Liabilities:	Accounts Payable	+amount (credit)

P&L: No impact at receipt stage
(expense recognized when parts are consumed)

Step 2: Stock Transfer (Main Warehouse to Workshop)

Purpose

Move received parts from the Main Warehouse to the Workshop Branch where the car is being serviced.

Trigger

- Warehouse manager initiates transfer after GRN is recorded
- Or automatic transfer when all PO items for a job are received

Endpoint

POST /api/company/{companyId}/inventory/transfer

Input

Field	Required	Description
fromLocationId	Yes	Main Warehouse location UUID
toLocationId	Yes	Workshop Branch location UUID
items[].productId	Yes	Product/part UUID
items[].quantity	Yes	Quantity to transfer
transferNote	No	Reference note

Process

1. Validate source location has sufficient stock
2. Create OUT movement at source location

- 3. Create IN movement at destination location
- 4. Decrease stock at source location
- 5. Increase stock at destination location
- 6. Create internal transfer journal (optional)

Database Changes

inventory_movements (OUT):

Column	Value
source_type	transfer
movement_type	out
quantity	Transfer qty
location_id	Main Warehouse (source)
transfer_reference	Transfer note/ID

inventory_movements (IN):

Column	Value
source_type	transfer
movement_type	in
quantity	Transfer qty
location_id	Workshop Branch (destination)
transfer_reference	Transfer note/ID

inventory_stock:

Location	Change
Main Warehouse	quantity -= transfer qty
Workshop Branch	quantity += transfer qty

Cases

Case	Behavior	Main Warehouse	Workshop
Transfer full qty (5 of 5)	All stock moves	qty = 0	qty = 5
Transfer partial qty (3 of 5)	Split stock	qty = 2	qty = 3
Transfer to multiple workshops	Multiple transfers	qty decreases per transfer	Each branch gets its share
Insufficient stock	BLOCKED	No change	No change

Case	Behavior	Main Warehouse	Workshop
Transfer back (return to warehouse)	Reverse direction	qty += returned	qty -= returned
Same location (source = dest)	BLOCKED	No change	No change

Accounting Impact (Optional Internal Transfer)

If tracked at cost center level:

Debit: Workshop Branch Inventory

Credit: Main Warehouse Inventory

(Asset-to-asset transfer, no P&L impact)

Step 3: Workshop Receives Parts

Purpose

Workshop branch confirms parts are physically available for the job.

What Happens

- Workshop inventory shows updated stock
- Job card prerequisites met (parts available)
- Technician can start work

Effects on Workflow

System Area	Change
Job Card	Parts readiness = Ready (unlocks "Start" action)
Car-In Dashboard	Parts column shows Received badge
Advisor Portal	Parts Order shows received count
Test Panel	Step 7 shows pass for goods received
Lead Stage	May advance to next stage if all parts ready

Configuration Required

Company Level

Setting	Description
<code>main_warehouse_location_id</code>	UUID of the default receiving warehouse location

Per Branch

Setting	Description
branch_location_id	UUID of the branch's inventory location

Setup Steps

- 1. Create inventory locations for Main Warehouse
- 2. Create inventory locations for each Workshop Branch
- 3. Set Main Warehouse as default receiving location in company settings
- 4. Link each branch to its inventory location

Complete Data Flow Diagram



Error Handling Summary

Error	HTTP	Message	Stage
PO not found	404	Purchase order not found	Receipt
PO cancelled	400	Cannot receive against cancelled PO	Receipt
Item not in PO	400	Item does not belong to this PO	Receipt
Over-delivery	400	Received qty exceeds ordered qty	Receipt
Already received	400	Item already fully received	Receipt
Insufficient stock	400	Not enough stock at source location	Transfer
Same location	400	Source and destination must be different	Transfer
Location not found	404	Inventory location not found	Transfer
No warehouse configured	400	Main warehouse location not configured	Receipt

Inventory Movement Types Summary

Movement Type	Source Type	Direction	When
in	receipt	Vendor → Main Warehouse	GRN recorded
out	transfer	Main Warehouse →	Transfer initiated
in	transfer	→ Workshop Branch	Transfer completed
out	consumption	Workshop →	Parts used in job card
in	return	Workshop → Main Warehouse	Unused parts returned

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